Fractions and decimals — Rounding · Multiplying and dividing · Recurring decimals · Mixed numbers

No calculator. Give fractions in their simplest form. ★ = challenge question.

A · DECIMALS

1 Round to 2 decimal places:
a 47.826 **b** 3.1049 **c** $0.096\,51$

> Look at the 3rd decimal place. **a** 47.83 **b** 3.10 (keep the zero to show 2 d.p.) **c** 0.10.

2 Round to 3 decimal places:
a 8.12345 **b** $0.045\,678$ **c** 219.9996

> **a** 8.123 **b** 0.046 **c** 220.000 — rounding up carries all the way through.

3 Use a written method to calculate **a** 3.4×2.6 **b** 14.5×12.3

> **a** $34 \times 26 = 884$; two decimal places in the question → 8.84.
> **b** $145 \times 123 = 17\,835$; two decimal places → 178.35.

4 Calculate **a** $0.5208 \div 0.004$ **b** $71.4 \div 0.03$

> Make the divisor a whole number by multiplying both numbers by the same power of 10. **a** $520.8 \div 4 = 130.2$. **b** $7140 \div 3 = 2380$.

5 Write each recurring decimal using dot notation:
a $0.5555\ldots$ **b** $0.454545\ldots$ **c** $0.217217\ldots$

> **a** $0.\dot{5}$ **b** $0.\dot{4}\dot{5}$ **c** $0.\dot{2}1\dot{7}$ — dots on the first and last digits of the repeating block.

6 Write each fraction as a decimal (use dot notation if it recurs):
a $\frac{4}{9}$ **b** $\frac{7}{12}$ **c** $\frac{3}{8}$ **d** $\frac{5}{11}$

> Divide top by bottom. **a** $0.\dot{4}$ **b** $0.58\dot{3}$ **c** 0.375 **d** $0.\dot{4}\dot{5}$.

B · FRACTIONS

7 Calculate
a $\frac{2}{5} \times \frac{10}{11}$ **b** $-\frac{3}{8} \times \frac{4}{9}$ **c** $\frac{7}{12} \times \left(-\frac{3}{14}\right)$ **d** $-\frac{5}{6} \times \left(-\frac{9}{10}\right)$

> Cancel first, then multiply; sign rules as for integers. **a** $\frac{4}{11}$ **b** $-\frac{1}{6}$ **c** $-\frac{1}{8}$ **d** $+\frac{3}{4}$.

8 Evaluate

a $\frac{1}{4} \div \frac{1}{3}$ **b** $\frac{3}{5} \div \frac{1}{10}$ **c** $-\frac{2}{3} \div \frac{5}{9}$ **d** $-\frac{7}{8} \div \left(-\frac{7}{12}\right)$

> Multiply by the reciprocal. **a** $\frac{1}{4} \times 3 = \frac{3}{4}$ **b** $\frac{3}{5} \times 10 = 6$ **c** $-\frac{2}{3} \times \frac{9}{5} = -\frac{6}{5} = -1\frac{1}{5}$ **d** $\frac{7}{8} \times \frac{12}{7} = \frac{3}{2} = 1\frac{1}{2}$.

9 Alex ran $1\frac{3}{4}$ miles, swam $\frac{2}{3}$ of a mile and cycled $4\frac{5}{6}$ miles. How far did he travel in total?

> Common denominator 12: $\frac{7}{4} + \frac{2}{3} + \frac{29}{6} = \frac{21}{12} + \frac{8}{12} + \frac{58}{12} = \frac{87}{12} = 7\frac{3}{12} = 7\frac{1}{4}$ miles.

10 Calculate

a $2\frac{3}{5} + 3\frac{1}{4}$ **b** $8\frac{2}{3} + 1\frac{5}{6}$ **c** $5\frac{1}{2} - 2\frac{4}{5}$ **d** $4\frac{1}{8} - 1\frac{5}{6}$

> **a** $5 + \frac{12}{20} + \frac{5}{20} = 5\frac{17}{20}$ **b** $9 + \frac{4}{6} + \frac{5}{6} = 9\frac{9}{6} = 10\frac{1}{2}$ **c** $\frac{11}{2} - \frac{14}{5} = \frac{55}{10} - \frac{28}{10} = \frac{27}{10} = 2\frac{7}{10}$ **d** $\frac{33}{8} - \frac{11}{6} = \frac{99}{24} - \frac{44}{24} = \frac{55}{24} = 2\frac{7}{24}$.

11 Evaluate, giving your answers in their simplest form: **a** $1\frac{1}{2} \times 2\frac{2}{5}$ **b** $2\frac{2}{3} \times 1\frac{1}{8}$

> Improper fractions first. **a** $\frac{3}{2} \times \frac{12}{5} = \frac{36}{10} = 3\frac{3}{5}$ **b** $\frac{8}{3} \times \frac{9}{8} = 3$.

12 Write each recurring decimal as a fraction: **a** $0.\dot{6}$ **b** $0.\dot{3}\dot{6}$ **c** $0.5\dot{2}$

> **a** $x = 0.666\ldots$, $10x = 6.666\ldots$; subtract: $9x = 6$, $x = \frac{2}{3}$.
>
> **b** $100x = 36.3636\ldots$; subtract x: $99x = 36$, $x = \frac{36}{99} = \frac{4}{11}$.
>
> **c** $10x = 5.222\ldots$ and $100x = 52.222\ldots$; subtract: $90x = 47$, $x = \frac{47}{90}$.

13 Find the side length of a cube with volume **a** $\frac{8}{125}$ m³ **b** $\frac{1}{64}$ cm³

> Side $= \sqrt[3]{\text{volume}}$: cube-root top and bottom. **a** $\frac{2}{5}$ m. **b** $\frac{1}{4}$ cm.

Check: count decimal places when multiplying · make the divisor whole when dividing · mixed numbers → improper fractions before × or ÷ · simplest form at the end.